AI and Inequality: The Full Revised Equation System

Top-Income Concentration as the Identified Channel

Joint Digital Transformation and the Middle-Income Trap — Equations (final revision)

June 15, 2026

What changed and why. The original framework routed inequality through the Gini ($-\eta_{10}\text{Gini}-\eta_{11}G\,\text{Gini}-\eta_{12}FG\,\text{Gini}$ in Eq. 14′). Empirically that channel is unidentifiable (Gini within/between = 0.07) and, where estimable, wrong-signed. Across six measures only the *tail-sensitive* ones detect an effect, and they detect it sharply. This revision rebuilds the inequality block around the top income share S_{it}^{top} and records the one empirical surprise: the skill-bias term *reverses sign* relative to the original theory — AI concentrates income at the top *more* where human capital is *high*, not low.

1 The identified inequality channel

Let G_{it} be AI intensity (within-year standardized), F_{it} FinTech, H_{it} human capital, and $S_{it}^{\text{top}} \in \{\text{top } 1\%, \text{top } 10\%, \text{Palma}\}$ the pre-tax national income share of the top group (WID). The law of motion for concentration is

$$S_{it}^{\text{top}} = \alpha + \gamma_1\, G_{it-1} + \gamma_2\, [G(1-H)]_{it-1} + \gamma_3 F_{it-1} + \Gamma' X_{it-1} + \mu_i + \lambda_t + \varepsilon_{it}, \tag{I1}$$

with the estimated signs $\gamma_1 > 0,\ \gamma_2 < 0$. The marginal effect of AI on concentration is

$$\frac{\partial S_{it}^{\text{top}}}{\partial G_{it}} = \gamma_1 + \gamma_2\,(1 - H_{it}) = \underbrace{(\gamma_1 + \gamma_2)}_{\approx 0} + |\gamma_2|\, H_{it}, \tag{I2}$$

which is *increasing in human capital*. This is the corrected mechanism: call it skill-complementary concentration — AI raises the top share most in skill- and capital-rich economies that can capture the returns, and negligibly where human capital is scarce. It is the opposite of the original skill-biased-displacement prediction ($\gamma_2 > 0$ on $G(1-H)$).

Dis-equalizing threshold. Setting (I2) to zero gives the human-capital level above which AI is dis-equalizing:

$$H^* = 1 + \frac{\gamma_1}{\gamma_2} \approx 0.06, \qquad \text{so AI raises } S^{\text{top}} \text{ for all but the lowest-}H \text{ economies, increasingly with } H. \tag{I3}$$

2 Macro complement and the dead channels

By the factor-income identity, a rising top share is the distributional face of income shifting from labor to capital, so the labor share $LS_{it} = 1 -$ capital share is the macro shadow of (I1):

$$LS_{it} = \alpha' - \theta_1 G_{it-1} - \theta_2 [G(1-H)]_{it-1} + \theta_3 F_{it-1} + \Gamma' X + \mu_i + \lambda_t + u_{it}. \tag{I4}$$

Prediction $\theta_1 > 0$ (AI lowers LS). Empirically θ_1 has the right sign but is *insignificant*: the labor share is too aggregated to resolve a top-tail movement. The Gini channel (Eq. $14'$) is dropped entirely — it is frozen within country and cannot enter a fixed-effects model.

3 Growth side: structural conditioning (unchanged, retained)

Inequality of *structure* (the dual economy) still conditions the growth complementarity, through a separate, identified channel. With $D^{\text{struct}}_{it} = \sigma^{KIS}_{it}(1 - H_{it})$,

$$g_{it} = \alpha + \beta_1 F_{it-1} + \beta_2 G_{it-1} + \beta_3 (FG)_{it-1} + \beta_4 \left(FG \times D^{\text{struct}}\right)_{it-1} + \Gamma' X + \mu_i + \lambda_t + \epsilon_{it}. \tag{G1}$$

So the two faces of inequality enter through different objects: *structural* dualism conditions the growth payoff (G1); *distributional* concentration is the AI-driven outcome (I1).

4 Forecast: the inequality AI may cause

Projecting (I1) forward gives the concentration path under a digital scenario s with an effect that matures along a GPT diffusion (J-)curve $m(t) \in [0, 1]$:

$$S^{\text{top}}_{i,t+1} = S^{\text{top}}_{i,t} + \kappa \Big[\gamma_1 G^s_{it} + \gamma_2 G^s_{it}(1 - H_i)\Big]\, m(t) + \eta_{it}, \tag{F1}$$

$$m(t) = \frac{1}{1 + e^{-k(t - t_0)}}, \qquad G^s_{i,t+1} = G^s_{it} + \rho_s\, \delta_G\, (1-d)^{\,t - t_0}, \tag{F2}$$

with $\rho_s \in \{0, 1, 2.2\}$ for Stall/Baseline/Joint-push and η_{it} a country growth shock. Both parameter uncertainty (draws of γ_1, γ_2) and process uncertainty (η) are propagated into fan charts. Higher S^{top} = more concentration.

5 Estimates on the assembled panel

Two-way fixed effects, country-clustered SEs, lagged regressors; income shares from WID, 130 countries, 2000–2024.

Forecast (F1) on the top-10% share: relative to Stall, a Joint-push raises the projected median middle-income top-10% share by ≈ 0.7 pp by 2040, ≈ 0.9 pp if the effect matures.

Identification caveat. (I1) is a within-country *association*; AI intensity is plausibly endogenous to concentration. The claim is "AI adoption is associated with rising top-income concentration," pending an external instrument (e.g. submarine-cable timing) for a causal reading.

Outcome	γ_1 (AI)	p	γ_2 ($G(1-H)$)	p	verdict
Gini (disp.)	-0.008	0.62	-0.001	0.95	tail-deaf: null
Labor share LS	-0.010	0.07	$+0.015$	0.06	macro: right sign, weak
Top 1% share	$+0.045$	0.08	-0.042	0.12	concentration ↑
Top 10% share	$+0.067$	0.008	-0.073	0.01	**concentration** ↑↑
Palma	$+1.52$	0.03	-1.51	0.04	concentration ↑

Table 1: The inequality effect of AI is invisible to the Gini and the labor share but sharp on the top shares: $\gamma_1 > 0$ (AI raises concentration), $\gamma_2 < 0$ (effect strengthens with human capital, per I2–I3). Between-country estimates are null, so identification is within-country over time.